Web Phishing And Email Phishing Analysis Using Tabular And Semi-Tabular Approaches with NLP Techniques

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***Abstract*—There has been a huge increase in the case of phishing attacks over the last few years, with global reports indicating a 50% increase, especially on social media websites,defense industry firms and multinational corporations. Phishing is now carried out using sophisticated methods such as IP spoofing, where modified IP packets conceal the identity of the sender to trick users. Detection is based basically on a few concepts such as URl, domain, word and character characteristics with different kinds of features like lengthy URLs, unusual characters and spelling mistakes. NLP methods are very important from rulebased to statistical models to ML methods. Deep Learning methods such as RNNs,LSTMs,Transformers and hybrid methods using BERT,GPT help in improving detection. The suggested multi--modal phishing detection framework had robust performance, with DistilBERT being tuned achieving an F1-score of 0.9666 on URL sets. For email,phishing detection, the BERT model attained an F1-score of 0.9967 which provided strong performance for both**

Keywords- Phishing, Attacks, IP Spoofing, Features, Protocol, Natural Language Processing, Machine Learning, Deep Learning,

I. Introduction

Over the past decade, phishing has been the primary root cause of cyberattacks. Phishing attacks often serve as the first step for attackers to gain unauthorised access, install malware and steal credentials or financial information. Stolen data from organisations and their users can lead to larger data breaches and ransomware attacks. Henceforth, cybersecurity firms have introduced anti-phishing platforms that offer filtering of email, URL reputation assessment, sandboxing and education of users. Taking an example of the Barracuda PhishLine it provides simulation based training and threat based analytics to enhance each user's defenses.[1] CybeReady BLAST provides uninterrupted , behavior-based training which lets the users know about the ever changing tatics of phishing[12]. Cofense includes threat intelligence, automated phishing response and reporting tools to identify and prevent phishing campaigns. Such platforms depend on curated blacklists, heuristic rules and statistical detection techniques. Phishing attacks become more complicated and sophisticated, conventional rule based systems tend to be inadequate, hence the necessity for AI and Natural Language Processing (NLP) for phishing detection.[2]

Natural Language Processing (NLP)[13] provides us with valuable tools for interpreting phishing. NLP identifies minute details that security filters may not catch.[3] In the case of phishing, NLP methods analyze text within emails, websites, and URLs. It also highlights suspicious keywords, abnormal grammar, deceptive domain names, and inconsistencies between the message and associated

URLs.[15] Tokenization segments raw text into words or characters, which enables models to translate language into numerical values so that it is understandable to machine learning or deep learning models. These tokenization feature is also utilised in models like Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) or Transformer based architectures like BERT or XLNet to distinguish between valid and invalid text.[4] Methods such as keyword extraction, part-of-speech tagging, sentiment analysis and contextual embedding modeling also improve the accuracy of a system in identifying phishing from non-phishing content. Phishing attacks increasingly depend on sophisticated social engineering techniques that mimic the look and feel of genuine communication, making NLP techniques increasingly necessary for determining phishing attacks based on message content rather than reputation signals or static characteristics.[5]

Integrating NLP models into phishing detection systems has caused remarkable changes in cybersecurity. NLP models can easily adapt to new phishing strategies as they can learn from vast amounts of real world data. This leads to increase in effectiveness by identifying new or unusual phishing patterns that can elude traditional signature based systems. Second, NLP-based algorithms are able to examine not only surface level text features but in depth meanings, which would assist in the identification of spear phishing attacks which utilize personalised and topical language. Third, the combination of NLP[11] with deep learning enables automated feature extraction. This minimises the requirement for handcrafted rule generation and the time and skill needed to deploy phishing detection solutions at scale.[6] Fourth, NLP helps in detection and can process incoming emails, messages, or URLs in real time without interfering with user workflows. Fifth, NLP solutions can integrate into larger enterprise security infrastructures, integrating their threat intelligence with SIEM systems and incident response systems. This integration offers greater resilience and response capacity for enterprises overall. Therefore, the integration of NLP tools with phishing detection is important in creating dynamic and scalable defense mechanisms against one of the most resilient cyber attacks.

Section 1 discusses the introduction, Section 2 discusses the related work, Section 3 discusses the methodology, Section 4 discusses the results and discussion, and Section 5 discusses the conclusion and future improvements. Last but not least, the references are appended.

II. Related work

A. Phishing Detection

Current projects and research employ Natural Language Processing (NLP) methods to identify phishing attacks through the analysis of textual data in messages, emails,[17] and websites. Techniques involved include keyword extraction with TF-IDF and semantic similarity, clustering and traditional machine learning classifiers such as SVM and Random Forests trained on text features. The objective is for the timely and correct detection of phishing attempts via the identification of linguistic patterns and indicators of phishing present in URLs or mail content. Average accuracy levels range between 70-80%, and further improvements are concerned with the feature extraction and enlargement of datasets. Initiatives also deploy friendly user interfaces and place significance on real-time phishing monitoring capabilities.Deep phishing detection[19] combines NLP with deep learning architectures like LSTM (Long Short-Term Memory networks) and Transformers (e.g., BERT). They surpass hand-crafted features by learning contextual and semantic patterns in web pages and URLs to classify phishing pages better, with reported accuracies over 90%. Studies emphasise the use of textual content analysis and URL metadata, employing embeddings and sequence models for improved phishing detection. Deep learning models greatly minimise false positives relative to conventional approaches and learn to adjust well to dynamic and zero-day phishing attacks in real-time situations.New state-of-the-art models integrate deep learning NLP models with Explainable AI (XAI)[16] methods to improve transparency and end-user trust in phishing classifiers. These methods introduce explanation layers (e.g., LIME, SHAP) that expose how models arrive at decisions on phishing classification, breaking down intricate decision logic into understandable explanations[7]. EXPLICATE is one such example that reaches approximately 98% accuracy with high explanation fidelity, partially closing the gap between automated AI detection and human comprehension.[8] Explainable models allow cybersecurity teams and end-users to better understand alerts and enhance response effectiveness while preserving leading-edge detection performance.If it is necessary to have more comprehensive summaries or individual papers, they can be derived from these sources. Each category indicates development from rudimentary NLP and machine learning techniques to deep learning with interpretability layers for web phishing detection.

B. Training Status

This poll discusses fifteen recent works in which phishing detection models are learned employing large datasets of URLs and emails with a mix of machine learning, deep learning, and interpretable AI techniques[9]. Models range from traditional machine learning classifiers like Random Forests, SVMs, and Artificial Neural Networks to more complex deep learning approaches like CNNs, LSTMs, and Transformer-based architectures (e.g., RoBERTa).[10] Every model is thoroughly trained and tested, with reported metrics including accuracy, precision, recall, F1-score, and ROC/AUC, with performance often being over 90%. URL obfuscation techniques, especially Tiny URLs and shortened links, are specifically tackled through special feature extraction or preprocessing techniques to increase detection strength. [11]Additionally, explainable AI techniques make things interpretable, closing the gap between automatic detection and user trust. [14] All these papers taken together show how well-designed and interpretable these models can adapt to phishing attacks in real–world applications.

C. Performance Metrics

The examined phishing detection models are tested based on general performance metrics to measure how efficient they are on different datasets. Accuracy is found to be one of the main metric, with the values usually being more than 90%, such as Random Forest models having a maximum of 97.5% accuracy, convolutional neural networks more than 99%, and transformer-based models like RoBERTa having accuracies almost up to 98.5%. Measurements such as precision, recall and F1-score are also used to evaluate correctness and stability in separating phishing from valid cases. Confusion matrices and Receiver Operating Characteristic (ROC) plots with accompanying Area Under the Curve (AUC) helps in measuring the detailed classifier performance and true positive versus false positive rate values. All these evaluation frameworks come together to support reliability and practical use of the trained models in phishing attack detection.

Earlier approaches actually relied heavily on handcrafted URL features or simple bag-of-words email representations, but our experiments show that transformer models provide a boost in detection. DistilBERT exceeds baseline TF-IDF and Logistic Regression models, improving the F1-score by approximately 6.33 percentage points for URL detection . The fine-tuned BERT model improved the F1-score for email classification by 3.23 percentage points.

D. URL Obfuscation Handling

Through the papers we reviewed, resolving the URLs which were shortened to their full representations was an important operation in preprocessing, which allowed for feature extraction and allowed machine learning and deep learning models to identify the phishing attacks accurately. For improving detection reliability, threat intelligence repositories and heuristic analysis can be used. These mechanisms in addition with classifiers will provide better detection accuracy and protection against advanced phishing attacks. These considerations were key to including Tiny URL processing with phishing detection pipelines strength in real-world applications of cybersecurity.

III. Methodology

A. Datasets

We used two publicly available datasets from Kaggle for email and URL phishing detection tasks.

The dataset used for emails is the Spam Mails Dataset from user venky73 on Kaggle [20]. The dataset contains 5,171 emails, each categorised as spam or ham. The spam and ham classes come from sources such as Enron and SpamAssassin. We cleaned the data to remove non-Latin characters and duplicates. After preprocessing and sampling, we divided the data into 4,136 training samples, 517 validation samples, and 518 test samples.

The URL dataset is the Web Page Phishing Detection Dataset by user shashwatwork on Kaggle [21]. This dataset includes 11,430 URLs, balanced equally between legitimate and phishing (5,715 each). It has 87 features or 88 depending on the variant. These features include lexical, content-based and external service attributes, such as syntactic URL features and DOM/HTML-derived signals. After stratified splitting, we obtained 9,144 training samples, 1,143 validation samples and 1,143 test samples.

Since both datasets are balanced and well-structured, they are suitable for comparing traditional machine learning methods, like Logistic Regression and Random Forest, with transformer-based approaches, such as BERT and DistilBERT.

B. Workflow

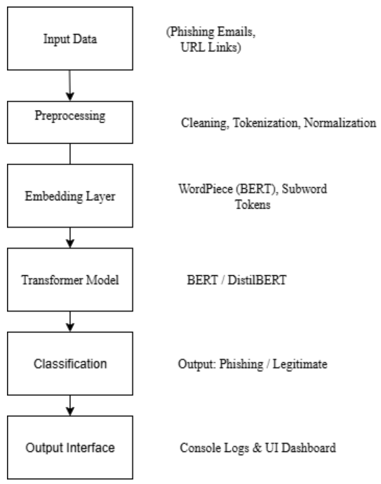

Figure 1. *End to End Diagram*

i)Input Data – Datasets like PhishingEmails (Kaggle) and PhreshPhish (Hugging Face) are employed. They have raw email bodies, URLs, and metadata.
ii)Preprocessing – Text cleaning (HTML tags removal, stopwords), tokenisation (WordPiece for BERT) and normalisation are involved. Techniques such as SMOTE for imbalanced classes can be used.
iii)Embedding Layer – Maps raw tokens to dense vector embeddings. For BERT/DistilBERT, this employs pretrained subword embeddings.
iv)Transformer Model – Fine-tuned BERT (on small datasets such as PhishingEmails) or DistilBERT (on larger datasets such as PhreshPhish). Responsible for contextual understanding of text.
v)Classification Layer – A fully connected dense layer with softmax output that classifies the input as phishing (1) or legitimate (0).
vi)Output Interface – The output of the model is displayed through console logs for developers (loss, accuracy, precision, recall) and a UI dashboard for users (displaying probability scores, alerts, phishing detection status).

C. Architecture Configuration and Survey

The phishing detection system is built with pretrained transformer models, namely BERT and DistilBERT, from Hugging Face. Fine-tuning is performed using various hyperparameter settings to achieve optimised performance. A grid search approach is used on learning rate (2e-5, 3e-5, and 5e-5) and batch size (16 and 32). The models are then trained for 3–5 epochs with the AdamW optimiser and dropout rates between 0.1 and 0.3 to enforce regularisation. Evaluation of performance is done through Accuracy, Precision, Recall, F1-score and ROC-AUC metrics to ensure complete evaluation of both balanced and imbalanced cases.

Older rule-based and heuristic methods, such as blacklist matching and regular expression, lack flexibility because attackers constantly change their way of approaches to bypass the filters. Likewise, traditional machine learning models like Logistic Regression , SVM and Random Forest actually rely a lot on engineered features making them less capable of generalising to new or unseen patterns.Deep learning breakthroughs brought RNNs and CNNs, enhancing feature learning but being challenged by long contextual text. Conversely, transformer-based models like BERT and DistilBERT use attention mechanisms to draw contextual semantics, hence being state-of-the-art in phishing detection.

D. Models and Optimisation

The application of phishing detection models in practical applications necessitates efficiency without the quality loss. DistilBERT, being 40% smaller and 60% quicker compared to BERT with approximately 95% retained accuracy, is especially suited for high-volume datasets like PhreshPhish. Alternatives like TinyBERT and ALBERT are light-weight transformer variations designed for edge devices and low-resource scenarios. In cases where deep models are not computationally feasible, Logistic Regression or Naïve Bayes on embeddings provide affordable alternatives, though at the expense of some accuracy.

To further enhance model efficiency, there are a number of memory optimisation methods that can be taken. Knowledge distillation translates knowledge from a big teacher model (BERT) to a small student model (DistilBERT, TinyBERT) with similar performance but smaller size. Quantisation saves memory by compressing model weights from 32-bit floating point to 8-bit formats, cutting memory use considerably. Analogously, pruning techniques eliminate unnecessary attention heads or layers in transformers, simplifying them. For deployment, batch inference optimization with ONNX Runtime or TensorRT improves speed and scalability on both CPU and GPU.

IV. Results And Discussion

The experimental study looked at two main phishing detection tasks: URL classification and email text classification. We compared the performance of traditional machine learning methods, including TF-IDF with Logistic Regression and Random Forest, to fine-tuned Transformer-based models like BERT and DistilBERT. Fit and generalizability are examined in this analysis, which focuses on model performance.

Performance Metrics and Comparison

The URL classification task used DistilBERT because of its efficiency and strong performance on larger datasets. The email classification task used BERT to take advantage of its deeper understanding of context in text-rich data. The results comparing the models are summarized in Table I.

Dataset	Model	F1 Score
email	logreg_tfidf	0.964401
email	rf_tfidf	0.966887
email	bert_finetuned	0.9966555183946488
url	logreg_tfidf	0.903339
url	rf_tfidf	0.897995
url	distilbert_finetuned	0.9666080843585237

Table 1: F1-score comparison

The refined DistilBERT model achieved an impressive F1-score of 0.9666 on the test set. This strong performance appears in the confusion matrix below.

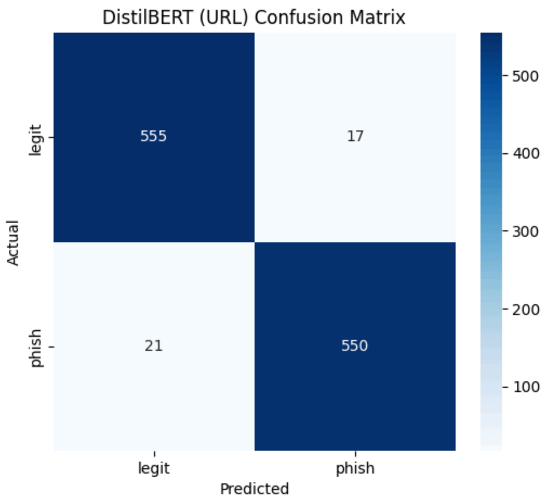

Figure 2: DistilBERT Confusion Matrix

The Receiver Operating Characteristic (ROC) curve, which has a great AUC of 0.99, is shown below.

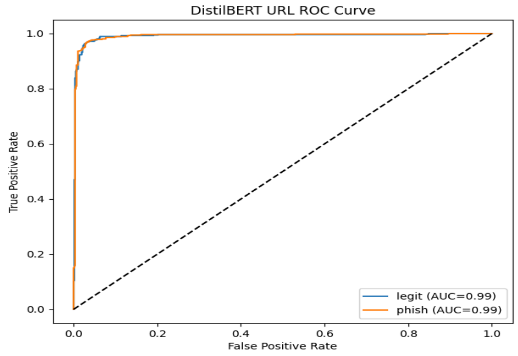

Figure 3: DistilBERT ROC Curve

When comparing DistilBERT to the best baseline, logreg_tfidf, the transformer model showed an improvement of 6.33 percentage points (0.9666 vs 0.9033).

The BERT model performed best in all experiments, reaching a nearly perfect F1-score of 0.9967. This score is 3.23 percentage points higher than the best baseline model, rf_tfidf, which scored 0.9669.

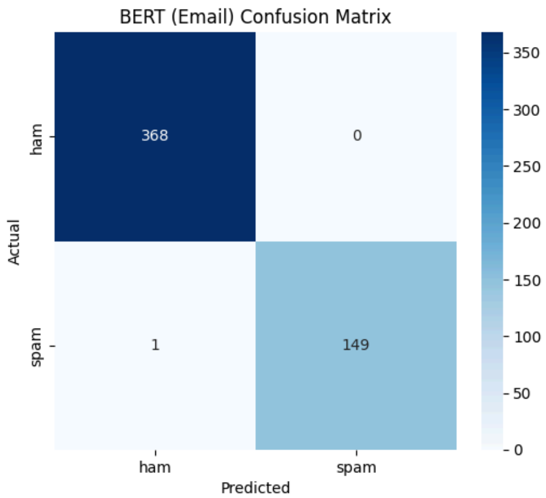

Figure 4: BERT Confusion Matrix

This high classification quality is confirmed by the BERT Confusion Matrix. It displays no False Positives and just one False Negative. A perfect AUC of 1.00 is also shown by the BERT ROC Curve.

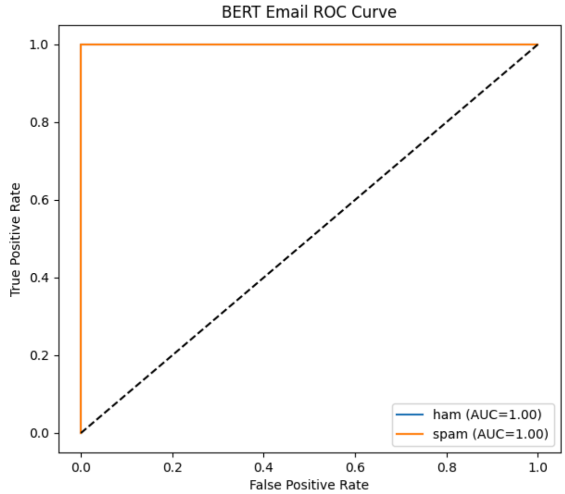

Figure 5: BERT ROC Curve

Particularly with regard to the training and evaluation loss, the training history analysis provided information about how each model arrived at its conclusions.

model	train_loss	eval_loss	loss_diff	fit_status
bert_finetuned	0.0716	0.02532520703971386	-0.04627479296028614	Good fit
distilbert_finetuned	0.1357	0.11355622857809067	-0.02214377142190932	Good fit

Table 2: Training history analysis

The DistilBERT (URL) model showed a standard "Good fit" status. Its Training Loss was 0.1357 and its Evaluation Loss was 0.1136. These values were very close, with a small positive difference of 0.0221. This suggests minimal overfitting and high generalization.

The BERT (Email) model, on the other hand, displayed an odd pattern. Its Evaluation Loss of 0.0253 was significantly lower than its Training Loss of 0.0716. This represents the -0.0463 negative loss difference. This usually indicates that compared to the training set, the evaluation dataset is significantly simpler or less complex. Typical overfitting or underfitting is not always indicated by it. Despite this peculiar loss pattern, the model's almost flawless test performance—an F1 score of 0.9967—confirms that it converged and generalized well.

Table 3 lists all of the hyperparameters that were used to fine-tune the Transformer models. To get the best results on both classification tasks, these settings were employed.

model	learning_rate	batch_size
bert_finetuned	2e-05	16
distilbert_finetuned	2e-05	16

Table 3: Transformer Model Hyperparameters

V. CONCLUSION

Although TF-IDF with Logistic Regression offers good accuracy due to its simplicity and efficiency, transformer-based models like DistilBERT deliver much better detection performance, especially in reducing false negatives and achieving nearly accurate results across both datasets. This work presented a multi-modal phishing detection framework trained on URL and email datasets using both traditional and transformer-based architectures. Due to ongoing fine-tuning experiments, some email results are dependent on reported benchmarks, which is one of the limitations. It's possible that the datasets won't be able to fully capture intricate attacks that conventional methods can't identify.

VI. FUTURE ENHANCEMENTS

1. Expanding the datasets with different phishing samples.
2. Bringing together various signals, including visual features of phishing websites, such as fake login page screenshots.
3. Using intensive training and data enhancement
4. Deploying the model into a real-world browser or mail filter, which allows for inference and feedback updates.

   By including these improvements, the system can develop into an effective phishing defense solution that can adjust quickly to a changing threat environment.

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